WRITING SAMPLE · POLICY ANALYSIS & PUBLIC FINANCE

Intergovernmental Hazard Mitigation

Reconciling climate change impacts with disaster relief: an analysis of federal hazard mitigation funding, intergovernmental finance and the case for proactive investment.

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Background

Climate change's impact on major disasters

As climate change increases the frequency, scale, severity and duration of disasters, the rate and number of persons displaced by disasters will likely increase.[1] Long-term recovery can minimize post-disaster disruption and address problems that existed before the disaster struck. The long-term recovery process reduces the disruption, economic dislocation and related social ills caused by a disaster.[2]

To best quantify the monetary amount of liability for climate change, we can look to previous litigation efforts and their legal outcomes. For example, one study found in 2016 that the rough estimate of potential liability for climate change for the Alliance of Small Island States was over $570 trillion. If emitters are held liable for the damages associated with climate change, this would create a powerful incentive for increased sustainability and mitigation efforts in both the public and private sectors.[3]

Governmental response to major disasters

Federal Emergency Management Agency (FEMA)

Although community recovery is primarily accomplished by the collective activities of individuals and community organizations, this study focuses on the roles of government intervention.[2] There was no overarching legislation or policy for disaster management at any level of government in the United States until after World War II. Established by President Jimmy Carter in 1979 through an executive order to consolidate federal disaster management programs dispersed across many agencies, the Federal Emergency Management Agency (FEMA) assumed most of the federal government duties specified in the various disaster relief acts passed since 1950.[2]

The Stafford Act

The primary law that now underlies the role of the federal government in presidentially declared disasters is the Robert T. Stafford Disaster Relief and Emergency Assistance Act.[2] The Stafford Act was established in 1988 to bring a systematic means of federal natural disaster assistance for state and local governments to provide aid to citizens.[4] The Act defines a major disaster as:

> "*any natural catastrophe (including any hurricane, tornado, storm, high water, winddriven water, tidal wave, tsunami, earthquake, volcanic eruption, landslide, mudslide, snowstorm, or drought), or, regardless of cause, any fire, flood, or explosion, in any part of the United States, which in the determination of the President causes damage of sufficient severity and magnitude to warrant major disaster assistance under this Act to supplement the efforts and available resources of States, local governments, and disaster relief organizations in alleviating the damage, loss, hardship, or suffering caused thereby.*"[4]

Ultimately, the Stafford Act gives FEMA the responsibility for coordinating government-wide relief efforts in response to a major disaster. However, FEMA does not exclusively direct all federal disaster response and recovery operations; the President also has the authority to

require any federal agency to use its resources in supporting state and local response and recovery efforts.[5]

Hazard mitigation efforts

State or local governments can, in principle, minimize losses over time by encouraging hazard mitigation.[6] Hazard mitigation planning aims to reduce the risk of loss of life and property by minimizing climate change-induced major natural disaster impacts.[5] FEMA sets several hazard mitigation planning requirements to best delegate emergency resources at the state, local and tribal levels. In preparing a state mitigation plan, planning initiatives at the regional level must be included to demonstrate integration with other planning initiatives and mitigation programs into ongoing state activities that achieve risk reduction and resilience. These plans follow a basic formula: organize the planning process and resources; assess risks; develop a mitigation strategy; and adopt and implement the plan.[7]

Eligibility for non-emergency disaster assistance grants is contingent upon hazard mitigation plan approval. While FEMA provides five grants eligible through an approved hazard mitigation plan, the most publicly available mitigation spending data are for the following three programs.[8]

The Pre-Disaster Mitigation Grant Program became the Building Resilient Infrastructure and Communities Program (BRIC) through an amendment in 2018. This program supports municipalities as they undertake hazard mitigation projects, reducing the risks they face from disasters and natural hazards.[9] While new funding mechanisms like BRIC have the potential to support increased investment in pre-disaster resiliency, the lack of permanent, predictable tools for pre-disaster preparedness and post-disaster recovery inherently limits states' and tribes' abilities.[1]

The Flood Mitigation Assistance Grant Program is a competitive grant that provides funding to municipalities for projects that reduce or eliminate the risk of repetitive flood damage to buildings insured by the National Flood Insurance Program.[10]

The Hazard Mitigation Grant Program (HMGP) offers funding for mitigation measures following a Presidential Major Disaster Declaration. A wide variety of mitigation projects qualify for funding, including utility retrofits, slope stabilization projects, acquisitions of hazard-prone homes and businesses, and developing and adopting required hazard mitigation plans.[11] An important distinction of the HMGP is that the locality need not be directly affected by a presidentially declared natural disaster; it must only reside within the affected state. The amount of funding is based on the total amount of disaster recovery funding approved for the affected state.[12]

Current state of hazard mitigation planning

The effectiveness of hazard mitigation planning has never been empirically contested. Between 1989 and 2009, the federal government spent over $96.1 billion on disaster assistance, and $4.4 billion was dedicated to hazard mitigation efforts. FEMA's largest source of post-disaster recovery funding is considered "permanent work"; examples of projects include repairs to roads and bridges, buildings and equipment, utilities, and parks and recreation. Of the $113 billion spent in response to the 2017 disasters, 0.08% of those funds were spent on preparedness efforts.[1]

In one report produced by The Pew Charitable Trusts, federal natural disaster mitigation spending is shown not to be comprehensively tracked, and very little data on state funding for these programs are publicly available. Even though federal grants save an average of $6 for every $1 spent on mitigation, most federal funding delegated to help states manage cost growth related to disasters is available only after an incident. Overall, the National Institute of Building Sciences found that mitigation efforts save up to $13 per $1 invested.[1,8,13–15]

On August 5, 2021, President Biden committed a historic $3.46 billion in hazard mitigation funds to reduce the effects of climate change. Every state, tribe and territory that received a major disaster declaration due to the COVID-19 pandemic became eligible to receive 4% of those disaster costs to invest in mitigation projects that reduce risks from natural disasters. Because of this investment, the program's funding available for declared disasters increased by 23% since the program began.[16]

Analyzing hazard mitigation

Rationales for government intervention

Climate change, the ultimate negative externality

Economic actors do not directly bear the climate change-related costs associated with the greenhouse gas emissions they can release into the atmosphere at no cost. Because market economies cannot efficiently manage open-access resources, greenhouse gas emissions produce a Pareto-inefficient outcome — rising temperatures — for actors outside of the private market. This market failure ultimately makes climate change a negative externality borne by a far larger population than the one responsible for the initial emissions.[17]

Disaster relief as a (bad) public good

Disaster relief can be considered a public good due to its nonrivalry and the geographic inability to exclude nonpayers; citizens tend to expect disaster protection as a part of civilized life. Governments hold a particular monopolistic position in this scenario because alternative providers are not available, and residents do not have the option of not paying for the government's service.[18]

Some argue that justifying federal provision of disaster relief as a public good should be avoided because the federal government does not possess a comparative advantage in the mass distribution of private goods. Emergency relief supplies, such as drinking water, meals and temporary housing, are private goods that, when consumed by one individual, reduce the amount available for all.[19]

Publicly funded disaster relief also unintentionally creates moral hazard. By shifting a large portion of the cost of relief to the public and lowering the cost of building homes and businesses in hazardous areas following devastating natural disasters, government intervention can incentivize others to relocate their homes and businesses to vulnerable coastal regions in the hope of receiving federal and state reconstruction assistance in the event of the next inevitable disaster.[19] Reasoned government intervention involves securing widespread coverage and regulating markets to ensure decision-makers feel the accurate cost implications of their choices.[18]

Intergovernmental finance and the budget process

Revenue sources: taxpayers to the federal government

The Federal Reserve Act of 1913 established the Federal Reserve System as the central bank for the United States. It works closely with the Department of the Treasury to manage the federal government's finances. Federal revenue consists primarily of individual, corporate and social insurance taxes collected from the people who live, work or do business in the U.S. each fiscal year.

Revenue sources: taxpayers to state government

The recession that began at the end of 2007 led many states to experience heightened levels of fiscal stress. State revenue growth either stagnated or, in some states, declined. It could be argued that the budgetary stress caused by the economic contraction has encouraged states to seek federal funds to help offset state disaster costs.[20]

Resources to provide services: the Disaster Relief Fund

The Disaster Relief Fund (DRF) is a congressional appropriation with which FEMA can direct, coordinate, manage and fund qualified response and recovery efforts for disaster emergencies in excess of state resources under the Stafford Act. In addition to funding hazard mitigation initiatives, the DRF also funds the repair and restoration of qualifying disaster-damaged public infrastructure, financial assistance to eligible disaster survivors, and Fire Management Assistance Grants for qualifying large forest or grassland wildfires.[21]

Funds are determined by the previous year's summary of allocations, obligations and expenditures for catastrophic events delineated by event, as well as the unobligated balance of funds carried over from the prior fiscal year. $24 million is then transferred to the Department of Homeland Security Office of Inspector General for audits and disaster investigations.[21]

Many factors can cause significant variability in disaster spending. One obvious issue with forecasting disaster activity and related costs is the inability to predict weather conditions, geological events, terrorist incidents and other scenarios related to DRF-funded disasters. A secondary issue arises when one of these events occurs and resources are diverted from existing mitigation and recovery efforts to address new, urgent and immediate concerns.[22]

From the Disaster Relief Fund to state governments: hazard mitigation assistance grants

Grants transfer spending power from one government to another, compensating governments for benefit spillovers to nonresidents, reducing the problems created by fiscal disparity, encouraging programs of exceptional national merit, reducing unique issues associated with regional economic decline, and encouraging governments to implement management reform by making it a condition for receiving aid.[18,23] Grants have also been used to provide states with funding to offset the costs of meeting federal regulatory standards or administering federal regulatory programs.[23]

The federal grant system succeeds in two objectives frequently cited by public finance experts: (1) encouraging states to use federal dollars to supplement rather than replace their spending on nationally important activities, and (2) targeting grant funding to states with greater programmatic needs and fewer fiscal resources. However, there are documented issues with the provision of federal grants to state governments. The first concerns the extent to which

federal grant dollars replace state dollars. The second concerns how federal dollars are distributed to balance differences among the states. The third concerns the ability to fund grant activities without federal assistance (state fiscal capacity).[23]

Studies show that federal grants replace a portion of state funds. These estimates suggest that a median of nearly 60 cents of every federal dollar is used to replace state and local funds that otherwise would have been spent on the aided activity; for every dollar of additional federal aid, states have withdrawn about 60 cents of their own spending.[23]

In theory, grants are meant to serve purposes beyond returning resources to taxpayers in the form of state services. Grants can also serve as a tool to encourage states to spend federal funds on nationally important activities for which they otherwise would have spent less. The amount of additional spending is affected by how federal grant funds supplement state funds; in effect, substitution allows a portion of federal grant funds to be spent on other state priorities.[23]

Because the recipient government receives all the political benefits associated with service delivery without having to raise the money, the risk of fund misallocation increases. To prevent misuse of its funds, the donor government seeks to ensure accountability in their use; these controls are often seen as barriers to an effective response by recipients.[18]

From state to local and regional governments

In program areas where states share financing responsibilities with the federal government, service levels depend to a significant extent on state fiscal capacities.[23] To manage their budget problems, states regularly "play the intergovernmental system" by transferring responsibilities to local governments.[18]

Cost sharing for labor, capital and supplies

State and local governments currently provide 25% of disaster costs on projects and grants to families and individuals, and the federal government assumes, at a minimum, 75% of all costs. There is no statutory limit on the number of people who can be helped following a disaster, nor a cap on the amount of federal funds expended. Some may argue that increasing the state share above 25% would be warranted given the federal government's fiscal condition.[20]

Provision of goods and services to the public

Government-sponsored disaster relief funds usually contain an inherent contradiction in providing low-cost insurance to high-risk properties,[6] and it is difficult to imagine that risk reduction would take place without risk-reflective premiums. Through the state delegation of hazard mitigation funds, many benefits present themselves to the public. For example, many states direct HMGP funds to the provision of generators for critical facilities or dedicate funds to strengthening dams near high-risk flood areas.

Ethics and professional values

The professionalism of emergency management

Many colleges and universities now offer degrees in emergency management and homeland security. This growth may be producing emergency managers who are more knowledgeable about the declaration process. Technological advances also enable state emergency managers to gather the information necessary; owing to these advances, modern state emergency managers are likely better able to document disaster losses than they were in the 1970s. This progression may have contributed to declining turndowns of requests for federal assistance, and state emergency managers may also have more confidence in advocating a request for a declaration.[20]

Cost-benefit analysis

Collaboration among federal programs, utilities, regulators, research institutions and communities will help ensure a positive return on investment for proactive hazard mitigation plans.[12] Because decision-makers may find it difficult to balance short-term recovery and long-term mitigation, it is important to better characterize and monetize the direct and indirect benefits of resilience investments. One broad way to show the increase in the frequency and magnitude of disasters is through the CPI-adjusted economic impact of those disasters.[12,24,25] An independent hazard mitigation council found in 2005 that FEMA mitigation grants are cost-effective and often lead to additional, non-federally funded mitigation activities.[13]

Historical costs

During the first nine months of 2021, there were 18 separate billion-dollar weather and climate disasters in the United States. Disaster costs for those nine months were $104.8 billion, already exceeding the cost of disasters for all of 2020 ($100.2 billion, adjusted for inflation to 2021). The total cost of the past five years ($691.7 billion) accounted for nearly a third of total disaster costs over the past 42 years (1980–2021), or $2,085 billion in inflation-adjusted 2021 dollars. This reflects a five-year average cost of almost $140.0 billion per year — a new record.

2021 broke the 16-event annual records of 2011 and 2017 and set a new annual record of 22 events. 2021 was the seventh consecutive year (2015–2021) with more than $10 billion in costs from weather and climate disasters in the United States. Over the past 42 years, the years with at least 10 separate billion-dollar disaster events include 1998, 2008, 2011–2013 and 2015–2021.[25]

Value of a statistical life (VSL)

Costs to society, such as casualties and displacement, are not easily measurable, so the cost-effectiveness of hazard mitigation is analyzed here with limited data. Because natural disasters directly cause casualties, the socio-economic costs and benefits can be calculated in part using the value of a statistical life (VSL). Boardman et al. suggest that, for policy purposes, the best point estimate of the VSL in the United States was around $11 million as of 2016.[26,27] Adjusted for inflation, $11 million in 2016 is equivalent to $3.78 million in 1980; this series was carried forward to 2021 and combined with forecasted billion-dollar events affecting the United States through 2029 (CPI-adjusted). Total cost is calculated by summing VSL losses and the costs of disaster recovery.

To estimate the potential benefits of hazard mitigation projects, the values of benefits and costs for the 2020–2029 period were projected using a linear forecast of all prior data points.[28,29]

According to the National Institute of Building Sciences' cost-effectiveness ratio of 13:1, the $3.46 billion recently dedicated to hazard mitigation would prevent an estimated $20.76 billion in eventual disaster recovery costs, providing a net benefit of $17.3 billion. FEMA's loss avoidance studies help quantify losses avoided — damage prevented due to the implementation of hazard mitigation projects — but they do not capture the socio-economic benefit of lives saved.

The potential yearly value of statistical lives lost varies by disaster type. This calculation would benefit from a social vulnerability assessment of the population's likelihood of living in a high-risk area, as well as the proportion of hazard mitigation funding going to projects that prevent damage from each disaster type. The number of lives that could be saved depends on the amount of mitigation funding dedicated to each hazard type; that information is beyond the scope of this study.

Environmental impact assessment

Because hazard mitigation needs are directly shaped by the effects of climate change, an environmental impact assessment (EIA) may be a viable complement to cost-benefit analysis that involves a more participatory decision-making process. Described as "a guide to better political judgment and performance," EIAs examine the potential effects of projects on the natural and social environment.[30,31] Nature-based hazard mitigation solutions, such as eco-systemic modeling, ecological engineering and blue/green infrastructure, would theoretically perform best under an EIA. However, research has found a multi-dimensional asymmetry between intentions and impacts in the outcomes of environmental policy design.[30,32]

Public positions and their rationales

Potential amendments to the Stafford Act

Potential amendments to the Stafford Act could limit the number of declarations issued or the amount of assistance provided. The Act could be amended so that federal assistance would only be available to states with corollary programs. Establishing these programs at the state level may increase state capacity to handle some incidents without federal assistance.[20]

Controlling declarations and their cost

Many argue that providing relief to disaster victims is an essential role of government, and that the increase in the number of declarations is justified because it is tied to increasingly severe weather. A similar argument could be made that the number of declarations should increase further, with adjustments to limit the amount of federal assistance granted in response. If the increase in declarations is a concern, perhaps because of the federal costs that accompany them, Congress may choose to address it through two approaches: (1) limiting the number of declarations, and (2) limiting the amount of spending after a declaration.[20]

Possible outcomes and solutions

The 2022 Hazard Mitigation Partners Workshop, an annual gathering of hazard mitigation stakeholders, called for discussion in the following areas.[33]

Future conditions, climate change and adaptation

Prioritizing the adaptation of electricity systems

Electricity systems are key to any state's economic and national security, yet they are vulnerable to many extreme weather events and natural disasters. Due to consistently high demand for this utility, outages and damaged infrastructure often cost billions of dollars in response and recovery.[12] Selected utility and transportation infrastructure case studies demonstrate a benefit-cost ratio of $4 saved for every $1 spent; an investment in a more resilient national energy system can therefore support the deployment of more innovative electric infrastructure.[12,15]

Widespread power interruptions impose many costs on customers and society. These costs fall into five categories: damage to utilities' infrastructure; interruptions to electricity customers, including lost production; indirect costs; intangible costs; and the costs of risk reduction and resilience enhancement, which are typically incurred before an interruption happens. Because very few studies have analyzed the economic impact of widespread power interruptions, these costs have not been considered in many policymakers' decision-making processes.[24,34]

Nature-based solutions

Ecological engineering and flood hazard mitigation

A 2020 study of nature-based solutions (NBS) for pluvial flood hazard mitigation took place in the Netherlands. Pluvial floods occur when surface runoff exceeds drainage capacity, mostly in urban regions during high-intensity, short-duration rainfall. Researchers found that simulated NBS were effective in flood risk mitigation, reducing both flooded area and flood depth; however, individual scenarios depended heavily on the location and extent of the NBS as well as storm intensity. NBS can be most effective for frequent rainfall events as well as for low-probability events, and the type of NBS determines implementation cost — water storage, for example, is more cost-efficient than green car parks and green roofs.[35]

Equity in hazard mitigation planning and projects

A rights-based approach to internally displaced persons

In 2019, researchers found that FEMA's cost-benefit calculations, while designed to reduce taxpayer risk, tended to exacerbate wealth inequality in urban areas affected by flooding. Similar studies suggest that if these disparate impacts continue, federal disaster aid will be more likely to help wealthy communities recover more quickly and with more resources than low-income communities and communities of color. Governments also have a particular obligation to protect against the displacement of groups with a special attachment to their lands, such as Indigenous peoples. To address this disparity, Perls recommends a rights-based approach to internally displaced persons focused on two fundamental rights: freedom from discrimination, and access to assistance for the special needs of vulnerable groups.[1]

Building codes and standards

Building codes as equitable preventive measures

Federal programs increasingly require the use of resilience-based codes and standards. Adherence to building codes is specified as a condition for a state's demonstration of commitment to a comprehensive mitigation program under 44 CFR §201.5(b)(4).[7] As it stands, 65% of counties, cities and towns across the United States have not adopted modern building codes such as the International Codes (I-Codes). In 2020, FEMA projected that $132 billion in property losses would be avoided between 2000 and 2040 if municipalities adopt the I-Codes, with counties facing high hazard levels, high growth and high need yielding the largest potential savings. Because residential buildings make up about 80% of buildings, adopting I-Codes in housing makes a substantial aggregate difference, and low-income housing built to I-Code standards reduces inequitable impacts on those least able to recover.[36]

Studies of hurricane damage led states to push for improved building codes, though some argue this emphasis undermines the importance of nature-based approaches.[37] In 2012, H.R. 3377 proposed increased hazard mitigation grants for states meeting minimum building codes.[38] For earthquake hazard mitigation, seismic hazard maps are typically used to develop building codes; these maps forecast levels of shaking based on assumptions about when and where earthquakes occur, their size, and the level of shaking in each region.[39]

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